CO_2-Based Physiological Disorder Detection Module for Oyster Mushroom Cultivation

Project Id: R26-ISE-002

Project Proposal Report

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3. Research objective – **Rack Level Micro-Climate Growth Optimizer for Oyster**

DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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13/03/2026

ABSTRACT

Oyster mushroom cultivation requires carefully controlled environmental conditions to ensure healthy growth and high yield. Among these conditions, carbon dioxide (CO_2) concentration plays a critical role in determining mushroom morphology and productivity. Excessive or unstable CO_2 levels can cause physiological disorders such as elongated stems, delayed fruiting, and pinning failure, which are often detected only after visible symptoms appear, leading to reduced productivity and economic losses. This study proposes a CO_2-Based Physiological Disorder Detection Module that continuously monitors carbon dioxide levels within mushroom cultivation racks and predicts potential stress conditions before visible symptoms occur. CO_2 sensors installed near the racks collect environmental readings at regular intervals and transmit them to a microcontroller for processing and analysis. The system introduces a CO_2 Stress Index (CSI), which converts raw sensor readings into a normalized indicator representing the stress level experienced by the mushrooms and categorizes conditions into low, medium, or high stress levels based on predefined thresholds. When abnormal patterns are detected, early warnings are generated to encourage farmers to inspect affected racks and take preventive actions such as improving ventilation. Additionally, historical data are stored and presented through graphical visualizations, enabling farmers to observe environmental trends and make informed decisions. By transforming sensor data into actionable insights, the proposed module supports early detection of physiological disorders and contributes to more efficient and sustainable oyster mushroom cultivation..

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INTRODUCTION

Oyster mushrooms are widely cultivated due to their nutritional value, short growth cycle, and relatively simple production process. However, successful cultivation depends heavily on maintaining appropriate environmental conditions such as temperature, humidity, and carbon dioxide levels. Among these factors, carbon dioxide plays a significant role in determining the development and morphology of mushroom fruiting bodies.

During mushroom growth, inadequate ventilation or poor environmental control can cause CO_2 levels to rise significantly within cultivation racks. Elevated carbon dioxide concentrations may lead to abnormal growth patterns including elongated stems, small caps, and failure of pin formation. Such physiological disorders negatively affect the quality and yield of the harvest.

In many small-scale and medium-scale farms, environmental monitoring is still performed manually. Farmers typically observe visible symptoms before taking corrective action. By the time these symptoms appear, the damage to the crop may already have occurred. Therefore, there is a need for a system capable of continuously monitoring environmental conditions and identifying potential risks before they become visible problems.

The advancement of Internet of Things (IoT) technologies has enabled the development of smart agricultural monitoring systems. Sensors can now collect environmental data in real time, allowing farmers to analyze conditions and respond quickly to potential threats. Applying these technologies to mushroom cultivation provides an opportunity to improve crop health and productivity.

This project proposes a **CO_2 monitoring and prediction module** designed specifically for oyster mushroom cultivation environments. The system collects CO_2 readings from sensors, analyzes environmental trends, and produces early warnings when abnormal conditions are detected. By providing timely alerts and visual insights, the system helps farmers maintain optimal growing conditions and reduce the risk of physiological disorders.

BACKGROUND AND LITERATURE REVIEW

Environmental monitoring has become a fundamental component of modern smart agriculture systems. The integration of Internet of Things (IoT) technologies allows continuous observation of environmental conditions such as temperature, humidity, and gas concentrations, enabling farmers to respond quickly to unfavorable changes and improve crop productivity [1].IoT-based monitoring systems typically consist of sensors, microcontrollers, communication modules, and cloud-based platforms that collect and analyze real-time data from agricultural environments [5].

Mushroom cultivation is highly sensitive to environmental conditions because mushrooms lack chlorophyll and depend entirely on controlled surroundings for growth. Oyster mushrooms (Pleurotus ostreatus) are widely cultivated due to their nutritional value, fast growth cycle, and relatively simple production requirements. However, their development is strongly influenced by factors such as temperature, humidity, light, and carbon dioxide concentration [2]. Among these, CO_2 plays a particularly significant role in determining the morphology of fruiting bodies.

Research indicates that elevated CO_2 levels during the fruiting stage can cause abnormal growth patterns, including elongated stems, reduced cap size, delayed fruiting, and failure of pin formation [2]. These physiological disorders reduce the market quality and yield of mushrooms, leading to economic losses for farmers. In poorly ventilated cultivation environments, CO_2 produced by mushroom respiration accumulates rapidly, especially in enclosed rack systems.

Recent advancements in IoT technologies have enabled the development of smart agricultural monitoring systems that collect environmental data in real time and present it through dashboards and alerts [5]. Microcontrollers such as the ESP32 are widely used in these systems due to their low cost, wireless connectivity, and ability to interface with multiple sensors [3]. Gas sensors such as the MQ-135 can measure CO_2 concentration and air quality parameters, making them suitable for agricultural applications [4]

Despite these advancements, many existing agricultural monitoring systems focus primarily on temperature and humidity monitoring, while gas monitoring — particularly CO_2 tracking in mushroom cultivation environments — receives comparatively less attention. Furthermore, most solutions present raw sensor data without providing predictive analysis or decision-support capabilities, requiring farmers to interpret environmental conditions manually before taking corrective actions. This limitation can delay responses to unfavorable conditions and increase the risk of crop stress [1][5].

Therefore, there is a growing need for intelligent monitoring systems that not only measure environmental parameters but also analyze trends and predict potential stress conditions. A system capable of converting CO_2 data into meaningful indicators of crop health can enable farmers to take preventive measures before physiological disorders develop. Such predictive approaches have the potential to improve productivity, reduce economic losses, and support sustainable mushroom cultivation practices by enabling proactive farm management [2][5].

RESEARCH GAP

Despite the growing adoption of IoT technologies in agriculture, several limitations still exist in current environmental monitoring systems used for mushroom cultivation.
First, many existing monitoring solutions focus mainly on displaying environmental data without performing predictive analysis. While these systems allow farmers to observe conditions in real time, they often fail to provide meaningful insights or early warnings about potential crop stress.
Second, available systems typically monitor parameters such as temperature and humidity, but carbon dioxide monitoring is often overlooked. However, CO_2 concentration is a crucial factor that directly affects mushroom morphology and fruiting processes.

Another limitation is the absence of rack-level monitoring. In mushroom farms where multiple racks are used for cultivation, environmental conditions may vary between racks due to ventilation differences. Most monitoring solutions provide only general environmental readings rather than localized measurements.

Furthermore, current systems rarely provide decision-support recommendations to farmers. Even when abnormal readings are detected, the system does not always guide the farmer toward the next appropriate action.

Therefore, there is a need for a monitoring solution that not only measures CO_2 levels but also analyzes trends, generates stress indicators, and provides early warnings for potential physiological disorders in oyster mushroom cultivation.

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Although several IoT-based agricultural monitoring systems have been developed, limitations remain in addressing CO_2-related physiological disorders in mushroom cultivation. Table 1 compares selected existing research with the proposed system.

Table 1:Research Gap Comparison

Feature / Capability	IoT Agriculture Monitoring [1][5]	CO_2 Growth Study [2]	Proposed System
Real-time monitoring	✓	✗	✓
Temperature & humidity monitoring	✓	✗	✓
CO_2 monitoring	Limited	✓	✓
Rack-level monitoring	✗	✗	✓
Predictive stress analysis	✗	✗	✓
Early warning alerts	✗	✗	✓
Decision support for farmers	✗	✗	✓
CO_2 Stress Index (CSI)	✗	✗	✓

Existing systems mainly provide environmental monitoring without predictive analysis or localized rack-level insights. While biological studies explain the impact of CO_2 on mushroom growth, they do not offer automated monitoring solutions. Therefore, this research addresses the gap by developing a predictive CO_2-based disorder detection module with early warning capabilities.

RESEARCH PROBLEM

Oyster mushroom cultivation environments can experience fluctuations in carbon dioxide levels due to poor ventilation, environmental changes, or improper farm management. When CO_2 concentrations remain high for extended periods, mushrooms may experience physiological stress that leads to growth abnormalities and reduced yield.

In many farms, these problems are detected only after visible symptoms appear. Manual monitoring methods are insufficient for identifying early environmental changes that may lead to crop stress. As a result, farmers often respond too late to prevent damage to the crop.

The lack of a predictive monitoring system capable of analyzing CO_2 levels and providing early alerts presents a significant challenge for mushroom farmers. Without timely information, it becomes difficult to maintain optimal environmental conditions and prevent physiological disorders. Therefore, the key problem addressed in this research is how to develop an automated monitoring system that can continuously track carbon dioxide levels, analyze environmental trends, and provide early warnings for potential physiological disorders in oyster mushroom cultivation

OBJECTIVES

Main Objective

To develop an automated CO_2-Based Physiological Disorder Detection Module that monitors environmental conditions in oyster mushroom cultivation racks, analyzes carbon dioxide levels, and provides early warnings of potential stress conditions. This will allow farmers to take proactive actions to prevent physiological disorders, improving yield, mushroom quality, and overall farm efficiency. The main focus is **predictive monitoring**, transforming raw sensor data into actionable insights for timely interventions.

Sub-Objectives

1. Measure CO_2 concentration in mushroom racks using environmental sensors.

   By deploying CO_2 sensors (e.g., MQ-135 or MH-Z19) near each mushroom rack, the system continuously records CO_2 levels in parts per million (ppm). This real-time measurement ensures accurate tracking of environmental conditions that directly impact mushroom morphology and growth.

2. Continuously collect environmental data at fixed time intervals.

   The microcontroller (ESP32/Arduino) collects readings every 5–10 minutes, storing and transmitting data to the backend server. This ensures no sudden changes in CO_2 levels go unnoticed, providing a continuous dataset for trend analysis and stress prediction.

3. Develop a CO_2 Stress Index (CSI) to represent stress levels affecting mushroom growth.

   The system converts raw CO_2 readings into a normalized CO_2 Stress Index ranging from 0 to 1. Low, medium, and high stress categories are defined, enabling clear i.nterpretation of mushroom stress levels and supporting informed decision-making by the farmer

4. Identify abnormal CO_2 patterns that may lead to physiological disorders.

   Using trend analysis, the system detects sustained high CO_2 levels or sudden spikes that indicate potential risk for disorders such as elongated stems, pinning failure, or delayed fruiting. Early detection allows intervention before visible damage occurs.

5. Provide early warning notifications for farmers when stress conditions are detected.

   When the CSI crosses predefined thresholds, the system automatically generates alerts (Low / Medium / High stress) and follow-up suggestions, guiding farmers to inspect specific racks and adjust environmental controls, such as ventilation or airflow.

6. Visualize CO_2 trends and environmental patterns through graphical dashboards

   Historical CO_2 data is presented in easy-to-read graphs, showing trends for each rack over time. These visualizations help farmers monitor environmental patterns, compare rack performance, and make proactive management decisions for consistent mushroom growth

System Architecture

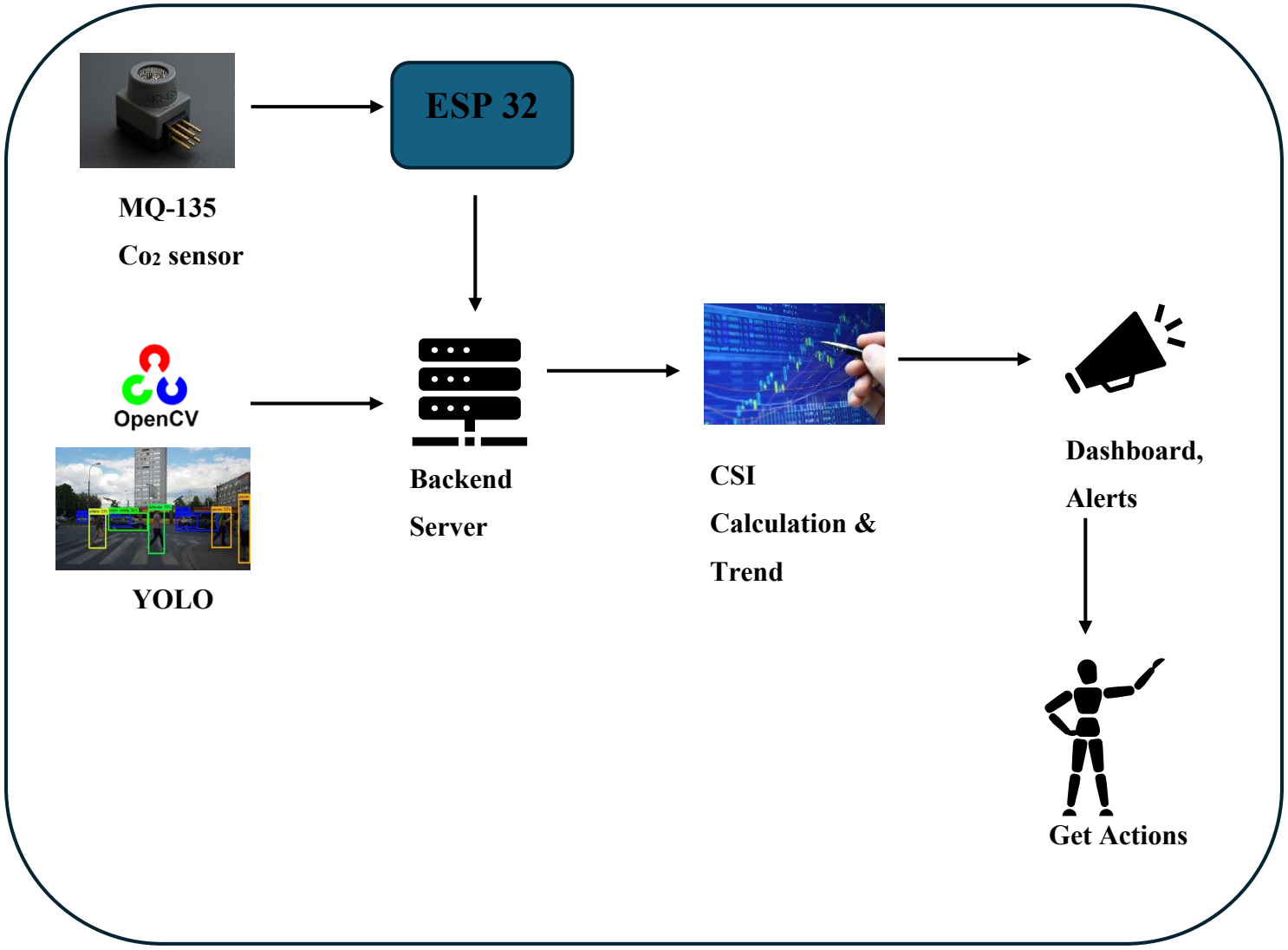


Table 2:System Architecture

The proposed system architecture consists of several interconnected components that enable continuous monitoring and analysis of environmental conditions in mushroom cultivation racks. CO_2 sensors (such as MQ-135) are installed near each rack to measure carbon dioxide concentration in parts per million (ppm). These sensors transmit readings to an ESP32 microcontroller, which performs initial data processing and forwards the information to the backend server through wireless communication.

At the backend, the system calculates the CO_2 Stress Index (CSI) using predefined thresholds and trend analysis techniques. The processed data are stored in a database for historical analysis. A dashboard interface displays real-time CO_2 levels, stress indicators, and environmental trends, allowing farmers to monitor

METHODOLOGY

1. Requirement Gathering

Information regarding oyster mushroom cultivation conditions is collected through an extensive literature review and consultations with experienced mushroom growers. This stage focuses on identifying the critical environmental factors that influence mushroom growth, morphology, and yield, with particular emphasis on carbon dioxide concentration. Functional and non-functional requirements of the proposed monitoring system are also defined based on real farm needs and existing technological limitations.

2. System Design

The system architecture is designed to support continuous, real-time environmental monitoring at rack level. The design integrates sensing components, a microcontroller unit, communication modules, a backend processing system, and a user interface. Emphasis is placed on scalability, reliability, and the ability to monitor multiple racks independently while maintaining centralized data analysis.

3. Sensor Configuration

CO_2 sensors are installed within mushroom cultivation racks to measure carbon dioxide concentration in parts per million (ppm). Sensors are positioned strategically near the growth area to capture localized environmental conditions accurately. Readings are recorded at fixed time intervals to ensure continuous monitoring and to detect sudden fluctuations in CO_2 levels.

4. Data Processing

Sensor readings are transmitted wirelessly to the backend system, where the data are filtered, validated, and stored in a structured format. The system computes a CO_2 Stress Index (CSI) using

predefined thresholds and analytical rules to represent the level of environmental stress affecting mushroom growth. Trend analysis techniques are applied to identify abnormal patterns such as sustained high concentrations or rapid spikes.

5. Visualization

Processed environmental data and stress indicators are presented through a graphical dashboard interface. Real-time charts and historical trend graphs display variations in CO_2 levels for each rack, enabling farmers to monitor environmental conditions easily and identify patterns that may affect mushroom development.

6. Alert Generation

When the CO_2 Stress Index exceeds predefined safe limits, the system automatically generates alerts indicating low, medium, or high stress levels. Notifications are designed to prompt farmers to inspect affected racks and take corrective actions such as improving ventilation or adjusting environmental controls.

7. Testing and Validation

The proposed system is evaluated under controlled or simulated mushroom cultivation conditions. Testing focuses on sensor accuracy, data transmission reliability, correctness of CSI calculations, responsiveness of alerts, and overall system performance. The effectiveness of the module in

PROJECT REQUIREMENT

Functional Requirements

1. **Measure carbon dioxide concentration in mushroom cultivation racks**
    - The system must accurately measure CO_2 levels in real time using sensors installed near each mushroom rack. This ensures environmental conditions critical to mushroom growth are continuously monitored.
2. **Record environmental data with time stamps for analysis**
    - All sensor readings are logged with precise timestamps and rack identifiers, creating a structured dataset for historical analysis and trend monitoring. This allows farmers to understand environmental patterns over time.
3. **Calculate CO_2 Stress Index based on collected readings**
    - Raw CO_2 readings are converted into a normalized CO_2 Stress Index (CSI) ranging from 0 to 1. This index represents the stress level experienced by the mushrooms, providing a clear, actionable measure of potential physiological disorder risk.
4. **Provide alerts when abnormal CO_2 levels are detected**
    - When the CSI exceeds predefined thresholds, the system generates notifications indicating Low, Medium, or High stress. Alerts are accompanied by suggested actions for corrective measures, enabling farmers to intervene before visible crop damage occurs.
5. **Display environmental trends and monitoring data through dashboards**
    - Real-time and historical data are visualized in easy-to-read dashboards with graphs and trends for each rack. This helps farmers make informed decisions, compare rack performance, and optimize cultivation conditions.

Non-Functional Requirements

1. **Continuous monitoring with minimal interruption**

- The system must operate reliably 24/7, ensuring no gaps in environmental monitoring that could allow stress conditions to go undetected.

2. **Accurate sensor readings for reliable analysis**

- Sensor precision and calibration are critical to ensure that the CSI reflects the true environmental stress levels experienced by the mushrooms.

3. **Efficient data processing and storage**

- Data must be processed quickly to calculate the CSI in real time, while historical records are stored efficiently for trend analysis and reporting.

4. **User-friendly dashboard interface for farmers**

- Dashboards should be intuitive, easy to navigate, and display information clearly, so farmers can understand stress levels and trends without technical expertise.

5. **Scalable system capable of supporting multiple racks**

- The architecture should allow monitoring of any number of racks, making it adaptable for small-scale and large-scale farms. Adding new racks should require minimal configuration.

BUDGET DETAILS

1. **Hardware Costs**

Item	Estimated Cost (LKR)
ESP32 Dev Kit (x1) / CO_2 sensor	10,500
Sensors integration accessories (wires, adapters)	1,500
Power supply / USB cable	1,000
Subtotal	13,500

Table 3:Hardware Cost

2. **Testing & Maintains**

Item / Description	Estimated Cost (LKR)
Software testing tools (Node-RED, React.js setup, debugging utilities)	2,000
Network testing (bandwidth simulation tools)	1,500
Subtotal	3,500

Table 4:Training & maintains

3. **Other costs**

Item / Description	Estimated Cost (LKR)
Documentation, printing, and presentation prep	1,000
Contingency (unexpected expenses)	1,500
Subtotal	2,500

Table 5:Other Costs

WORK BREAKDOWN STRUCTURE

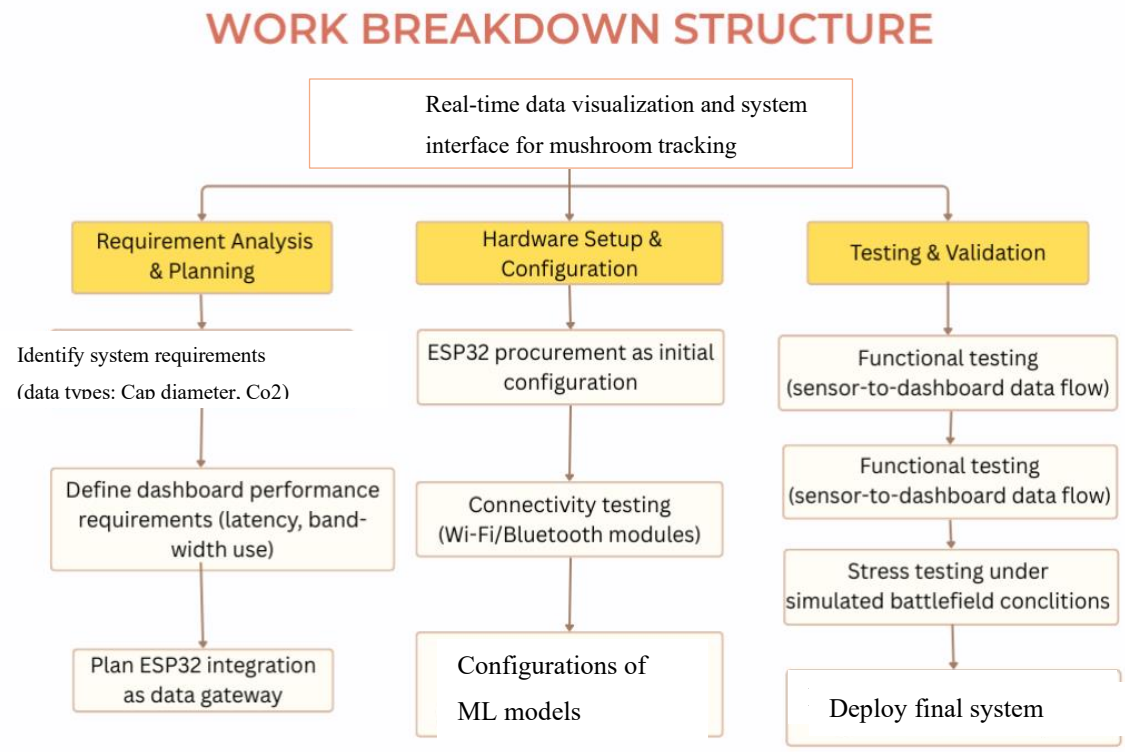


Table 6:Work Breakdown Structure

TIMELINE

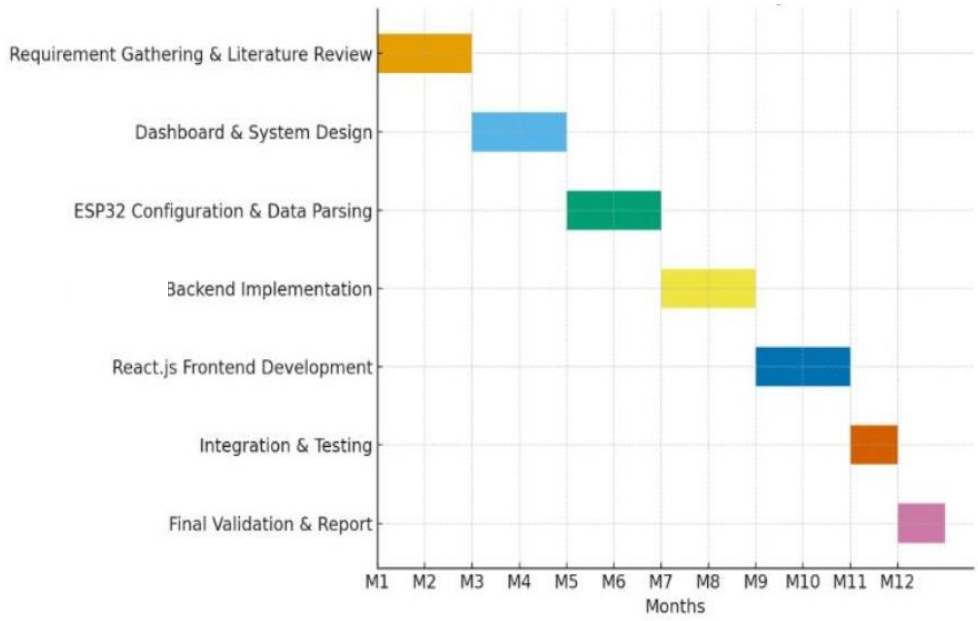


Table 7:Gantt Chart

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